

ANKER AI

Scientific Thesis

Scoring, Matching, and Extraction Methodology for AI-Native Investor Intelligence

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Version: Versioned with codebase. This document reflects current implementation; see [lib/ai/*](#) git history as ground truth.

The principles, evidence requirements, and decision rules behind every scoring, matching, and extraction choice in the platform. If a feature conflicts with this document, the document wins and the feature is the bug.

Abstract

Private capital allocation is a search problem under heavy information asymmetry. Three populations sit on each side of every match: funds raising from LPs and family offices, founders raising from VCs and angels, and investors screening managers or deals. Each transaction carries a non-trivial qualification cost — deck review, reference calls, data-room diligence, DDQ. The default state is mutual information starvation: neither side knows enough to qualify the other without a meeting, but a meeting itself costs more than the expected value of any single random match.

Anker's architecture addresses this by dropping the qualification cost per match until matches that previously were not worth running become net-positive in expected value. This is achieved through structured, source-anchored intelligence extraction — making computable the unstructured documents (pitch decks, fund materials, LP DDQs) that represent each actor in the market — and a transparent, additive scoring engine that ranks counterparties without producing opaque outputs.

This document formalises the evidence requirements, scoring methodology, extraction pipeline, and data quality posture that govern every Anker output. It is a living specification: when implementation diverges from this document, the document wins.

1. The Problem We Model

The fundamental inefficiency in private capital markets is not a lack of supply on either side of the table. It is the cost structure of the qualification process. Under the current default operating model:

- A fund manager building a \$20M emerging-manager raise must cold-contact 400–800 LPs to close 15–25 commitments.
- A founder raising a \$3M pre-seed round will send 200+ outreach messages to secure 30–40 first meetings.
- An LP or VC analyst screens 5–15 decks per week, spending 20–45 minutes on each before filtering 80–90% at the first touchpoint.

The root cause of this volume-to-conversion ratio is mutual information starvation. Both parties lack the structured intelligence to determine whether a match is worth running before the first contact.

Network effects patch this gap for well-connected incumbents but systematically disadvantage first-time fund managers, immigrant founders, and geographically distributed actors.

Anker models this as a search problem: given a structured profile of Actor A, find the N actors in the database whose structured profiles best satisfy A's mandate constraints across a defined set of weighted dimensions. The key insight is that the qualification work must happen before the email, not during the meeting.

2. Decision Principles

The following rules govern every scoring, extraction, and matching module. They are stated plainly because their purpose is operational, not rhetorical.

2.1. Evidence Before Inference

Every non-null field returned by an extractor must be supported by a phrase from the source document. If a field cannot be cited, it returns null. Plausible-looking guesses are categorically excluded: the cost of a wrong fact downstream (an incorrect DPI figure, a misidentified fund stage) is orders of magnitude higher than the cost of a missing field.

This principle is operationalised in the extractor system prompts:

RULE *FILENAME IS NOT EVIDENCE. The document filename tells you nothing about content. Do not expand acronyms, infer dates, or guess the firm name from the filename.*

RULE *CONFIDENCE MUST REFLECT EVIDENCE. If you returned mostly nulls because the text is sparse or image-heavy, set confidence to 0.1–0.3.*

Implementation: enforced in lib/ai/fund-deck-extractor.ts, lib/ai/fund-deck-analyzer.ts, lib/ai/pitch-deck-analyzer.ts, and lib/matching/v2/document-extractor.ts. When pdf-parse yields sparse text, pages are rendered to PNG and processed through Qwen-VL-OCR before falling back to nulls. OCR-sourced content carries a [source: Qwen-VL-OCR pass ...] tag, allowing the model to treat it as evidence at lower trust than direct PDF text extraction.

2.2. Source-Anchored Scoring

Every numeric score — LP-fit score, investor match score, deck critique score — is computed from a small number of explicit signals with documented weights. No single opaque score is surfaced without a breakdown. Every result page exposes the dimensions, the per-dimension score, and a one-sentence justification per dimension.

The fund-deck LP critique uses six lenses with weights that vary by manager maturity:

Lens	Emerging Manager Weight	Established Manager Weight
GP background & track record	1.0×	1.5×
Sector focus & differentiation	1.4×	1.0×
Market analysis & TAM rigour	1.1×	1.0×
Thesis alignment to 2025–26 trends	1.5×	1.2×

Fund unit economics	1.2×	1.2×
Claims verification	1.4×	1.5×

Rationale: *The emerging-manager profile up-weights thesis and sector edge because those are the dimensions where a new GP can plausibly outperform an established one. The established profile up-weights track record and claims verification because LPs allocating to a Fund V should expect institutional-grade reporting.*

2.3. Claims Verification as a First-Class Output

The fund-deck analyzer returns an explicit claimsReviewed[] array. Each item is structured as:

```
{ claim, category, verifiable, status, comment }
```

Categories: track_record, market_size, team_credential, portfolio_outcome, thesis_evidence, other. The verifiable field classifies each claim as:

- public — can be independently confirmed against external records
- private_audit — requires DDQ or LP reference call to verify
- unverifiable — cannot be checked against any external record

The status field is one of: plausible, suspicious, or contradicted. This is the single highest-leverage output of the analyzer. LPs prioritise follow-up calls based on the suspicious rows. Emerging-manager GPs use it as a pre-pitch self-audit checklist before submitting to LPs.

2.4. Confidence Is Honest

Confidence reflects source quality, not model certainty. A sparse image-only deck that was successfully OCR'd scores approximately 0.85, because OCR introduces a small uncertainty layer over direct text extraction. A confident-looking model output sourced from the filename alone scores 0.1. The UI surfaces confidence as a numeric badge alongside every extraction result; any value below 0.5 triggers a human-review prompt before the output can inform an outreach decision.

2.5. Failover, Never Fall to Silence

The AI provider chain (Anthropic → OpenAI → Gemini → Qwen → Mistral) fails over automatically on 429 or 5xx responses. When every provider fails, the extractor returns a heuristic fallback marked with confidence ≤ 0.3 and an explicit note explaining the failure mode. The UI displays the heuristic source prominently — a user must never confuse a fallback estimate with a verified extraction. The PDF vision dispatcher has four independent processing paths, ensuring the system continues operating even when one provider blocks a particular geography or document type. Qwen-VL-OCR serves as a fifth-line defence for image-only decks. The system never refuses to process a deck because of format.

3. Matching Methodology

3.1. LP Matching (Fund → LPs)

The matchmaker (lib/matching/v2/engine.ts) scores every firm in the database against a fund profile. The score is a linear combination of six explicitly weighted signals:

- Stage and sector alignment — does the LP's mandate match the fund's strategy?
- Geography fit — both HQ proximity and the LP's investing geography

- Cheque size match — average LP ticket size vs. the fund's minimum commit
- Right-sized portfolio — LP's prior fund commitment sizes relative to the fund's target raise
- Lesser-known multiplier — when the GP wants to avoid household-name LPs, the long tail receives elevated weight
- Thesis keyword overlap — semantic match between the fund's thesis and the LP's stated focus, computed via pgvector cosine similarity when both sides have embeddings

Output tiering: Tier 1 ≥ 9.0 , Tier 2 ≥ 7.0 , Tier 3 ≥ 5.0 , Tier 4 ≥ 3.0 , Drop < 3.0 . For a \$20M emerging-manager fund-of-funds, Tier 1 + Tier 2 typically yields 30–80 qualified LPs. The surfaced list is capped at 200 for ergonomic review. Each result exposes which signal contributed which proportion of the score — no black box.

3.2. Investor Matching (Founder → VCs)

The founder-side engine uses the same architecture with a different weight profile: stage focus dominates, followed by sector, geography, and then cheque size. It re-uses the same `score_investors` tool exposed by the agentic assistant interface, ensuring the dashboard and the autonomous agent produce identical results from identical inputs.

3.3. Why No Machine-Learned Ranker (Yet)

The database currently contains approximately 18,000 firms and 47,000 individuals. A learned ranker requires labelled match outcomes — which introductions converted to meetings, which meetings led to term sheets, which closed. These labels are being collected via the outreach and pipeline tracking modules, but the corpus is well below the scale at which a learned ranker statistically outperforms the additive linear model.

The planned transition point is $\geq 10,000$ labelled outcomes across at least three distinct funds. Until that threshold is reached, the linear additive model is the correct choice on two grounds: it is debuggable (a GP can ask 'why is this LP Tier 1?' and receive a defensible answer), and it does not amplify biases already present in the training data the way a black-box ranker would.

4. Extraction Methodology

4.1. The Four-Step Pipeline

Every uploaded document passes through the following sequence:

1. Text extraction via `pdf-parse`. Returns text content, page count, and image-only page count.
2. Sparseness check. Document is classified as sparse if image-only pages exceed 50% of total pages OR extracted text is fewer than 200 characters.
3. OCR fallback for sparse documents. `lib/ai/pdf-ocr.ts` renders each page to PNG via `pdfjs-dist + @napi-rs/canvas`, then processes via `Qwen-VL-OCR`. Default limit: 15 pages.
4. Vision dispatch. Extracted or OCR'd text plus optional inline images are submitted to the active provider in the failover chain. All prompts include the page-source tag, enabling the model to differentiate between direct text, OCR evidence, and absent evidence.

This pipeline is shared across all PDF-handling routes: LP matchmaking extraction and analysis, find-investors deck critique, documents page deck analysis, and the autonomous assistant's `analyze_image` and `ocr_image` tools.

4.2. Why Qwen-VL-OCR Specifically

Three alternatives were evaluated:

- Tesseract: excluded because it requires a system binary not available in Vercel's Lambda runtime, and its OCR quality on stylised slide text is inadequate.
- Generic vision LLM (Claude, GPT-4o): viable but expensive per page, with no OCR-specific fine-tuning.
- Qwen-VL-OCR: free-tier model specifically fine-tuned for OCR, runs through the same OpenAI-compatible endpoint as the Qwen stack, and preserves structured text reading order with numbers and URLs intact.

Trade-off: sequential rate limits on the free tier make a 12-page deck take approximately 72 seconds. This is acceptable given that accurate extraction is worth the latency, and the 15-page cap keeps total processing within Vercel's maxDuration: 240 constraint for the analyze route.

4.3. What We Deliberately Do Not Do

The following inferences are categorically excluded from all extraction modules, regardless of how plausible they appear:

- We do not infer fund or founder names from filenames.
- We do not infer 'Fund I' from the absence of a fund number — we return null if the deck does not contain an explicit statement.
- We do not invent DPI, TVPI, or IRR numbers when no numeric value is visible in the document.
- We do not score a deck on dimensions where the underlying slide is not accessible to the extractor.

5. Data Quality Posture

5.1. Idempotent Ingest Scripts

Every curated dataset (scripts/ingests/*.sql) is idempotent: re-running the script skips rows already present. Firms deduplicate case-insensitively by name. Individuals deduplicate by lowercased email, falling back to (firm_id, full_name) when no email is present. This architecture enables a curator to re-export the same data source monthly and run the same script — only the diff lands in the database.

5.2. Cleanup as First-Class Operation

scripts/cleanup-investors-firms.sql removes the canonical garbage patterns that accumulate from heterogeneous data sources:

- Duplicate-token names (e.g. 'Cynthia Ringo Ringo' — a Folk export artefact)
- Names containing corrupt characters from the Unicode blocks: †¥Ł€◇⊙*φψξ⊕⊖
- Rows where first_name = last_name with no associated email
- Exact (name, email) duplicate pairs
- Empty or null firm names

The cleanup script is wrapped in BEGIN / ROLLBACK by default. The operator must explicitly flip the trailing line to COMMIT. Database backups are created within the same transaction.

5.3. Provenance via Source Field

Every investor row carries a source field tagging its origin. Representative values:

- 8fundraising-lp-drop-04 — 150 US family offices, curated list

- folk-european-family-offices-founders — 100 EU founders from Folk
- discover-enrichment — discovered and enriched via web crawl
- linkedin-scrape — scraped via the LinkedIn ingest agent

Provenance is enforced at insert time and is never overwritten by enrichment. This enables users to filter the database to 'rows from trusted curated lists' and use that subset as a high-precision seed for outreach campaigns.

6. Privacy and Consent Posture

The platform stores professional contact information for investors and firm representatives. The scope of stored data is strictly limited to:

- Public LinkedIn profiles and LinkedIn URLs
- Public website URLs and company email addresses
- Job titles and locations as reported in public professional bios

Personal data — private mobile numbers, home addresses, family relationships — is not stored even when present in source CSVs. Ingest scripts strip those columns before COPY.

Outreach rate limits are enforced by the platform defaults: 25 LinkedIn connection requests per day, 50 follow-up direct messages per day, weekday-only sends. Every outbound message is sent via Resend or the user's own Gmail OAuth account — the platform never sends from an address it does not control.

7. Planned Methodology Extensions

The following additions are scoped and will be documented here when implemented:

- Learned ranker for LP matching: to be introduced once $\geq 10,000$ labelled match outcomes are available across multiple fund closes. Will initially run in parallel with the linear model to validate before replacing it.
- Federated deduplication of firms across multiple ingests: using semantic similarity over pgvector embeddings to identify the same firm appearing under variant names across different source lists.
- Citation graph for newsroom: every article will cite sources stored as structured (source, year, url) tuples. Inline citations are already extracted; the next step is the cross-article reference graph.
- Audit-grade extraction logs: enabling a GP to replay exactly which provider, prompt version, and source page produced any given extracted field. Required for institutional-grade due diligence reporting.

This document is versioned alongside the codebase. The git history of lib/ai/* is the authoritative ground truth. Any principle stated here that conflicts with current implementation should be treated as a filed bug, not a known exception.

Appendix A: Key File References

File / Path	Purpose
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<code>lib/ai/fund-deck-extractor.ts</code>	Fund deck structured extraction — enforces evidence-before-inference
<code>lib/ai/fund-deck-analyzer.ts</code>	Fund deck LP critique — six-lens scoring + claims verification
<code>lib/ai/pitch-deck-analyzer.ts</code>	Founder pitch deck analysis — investor-lens critique
<code>lib/matching/v2/engine.ts</code>	Core matchmaking engine — linear additive scoring
<code>lib/matching/v2/document-extractor.ts</code>	Document extraction for matching pipeline
<code>lib/ai/pdf-ocr.ts</code>	OCR fallback — Qwen-VL-OCR + pdfjs-dist page rendering
<code>scripts/ingests/*.sql</code>	Idempotent ingest scripts for curated datasets
<code>scripts/cleanup-investors-firms.sql</code>	Canonical garbage removal from investor database